A Novel Mathematical model for the Spread Prediction in Alloy Bar Rolling by the Round-Oval Pass Sequence

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Abstract—**In alloy bar rolling by Round-Oval pass sequence, the critical point on the contact boundary is solved and the cross-section profile parameter of outgoing workpiece is represented, then the equivalent height of outgoing workpiece and mean roll radius at the exit section is derived to correct the spread formula. Based on the initial spread calculated by the modified Shinokura and Takai spread formula, an iterative model for the spread prediction in alloy bar rolling is built to improve the accuracy of spread prediction. The GCr15 bar rolling experiment has examined the validity of this model. Compared with experimental data, this iterative model is precise enough and can be used to determine the rolling parameter and design the profile of finishing groove.**

Keywords- Alloy steel; Bar rolling; Round-Oval-Round; Critical point; Spread Prediction; Iterative model

1. Introduction

In rod or bar rolling, the lateral flow of deformed workpiece is complicated. The spread is influenced by many factors, such as roll diameter, reduction ratio, and length of contact arc, the groove profile, the width of workpiece, the friction coefficient, and the material of workpiece. Some scholar studied the law of the lateral flow of deformed workpiece in the groove, and represented their empirical-theoretical analytical model on the base of the rolling experiments, such as Smirnov spread formula [1], Wusatowski spread formula [2], Jiujin and Keliwencuov spread formula [3]. This model was derived by rolling plain carbon steel in groove, moreover, the equivalent reduction and mean roll radius was not been determined exactly. So it can't be used directly to predict the spread of alloyed steel in groove.

2. The spread prediction model for alloy bar rolling

2.1 The initial spread model for alloy bar rolling

Shinokura and Takai[4-7] studied the process of round-oval-round pass rolling and proposed a spread formula for rolling plain carbon steel. Gubujin and Qixikov [8] studied the influence of the workpiece's material upon the spread, and found that the spread of alloy bar is 20~30% bigger than that of carbon steel bar. So the influence of the steel's component upon the spread was taken into account to modify the spread formula in alloy bar rolling, and the value of influence coefficient m was given. For calculating the spread during alloy bar rolling, Shinokura and Takai Spread Formula should be modified as follows:

$$\Delta b = m \cdot \gamma \frac{\sqrt{\overline{R_0} \cdot \left(\overline{H_0} - \overline{H_{m0}}\right)}}{W_0 + 0.5H_0} \frac{A_h}{A_0} \cdot W_0 \tag{1}$$

$$\overline{H_0} = \frac{A_0 - A_s}{2C_{Y0}} \tag{2}$$

$$\overline{H_{m0}} = \frac{A_0 - A_s - A_h}{2C_{Y0}} \tag{3}$$

$$\overline{R_0} = R_{mzx} + \frac{G}{2} - \frac{\overline{H_{m0}}}{2} \tag{4}$$

Where $\overline{H_{m0}}$ the cross-section effective height of outgoing workpiece, $\overline{H_0}$ the cross-section effective height of incoming workpiece , $\overline{R_0}$ the mean roll radius, W_0 is the maximum width of the incoming workpiece, H_0 the maximum height of the incoming workpiece, A_0 cross-section area of the incoming workpiece, A_s is the area of non-contact section, A_h is the reduction area along the direction of Z axis, and γ is a correction factor, which varies from 0.8 to 1.1. According to the research of Shinokura and Takai[4], γ is 0.97 in round-oval pass rolling, and γ is 0.83 in oval-round pass rolling. m is the influence coefficient of steel's component. For the plain carbon steel Q195, m is 1.0. For the pearlite-martensite type steel (for example GCr15, 4Cr13, 40Cr), m varies from 1.24 to 1.308).

In round-oval pass rolling (Fig.1) , the reference point $\left(C_{Y0}, C_{Z0}\right)$, A_h and A_s can be obtained by

$$C_{Z0} = \frac{R_1^2 - R_a^2 - D_Z^2}{2D_Z} \tag{5}$$

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$$C_{Y0} = \sqrt{R_a^2 - C_{Z0}^2} \tag{6}$$

$$D_Z = R_1 - H_p/2 \tag{7}$$

$$A_h/2 = \arctan\left(C_{Y0}/C_{Z0}\right) \cdot R_a^2 - \left[\int_{-C_{Y0}}^{C_{Y0}} \left(\sqrt{R_1^2 - y^2} - D_Z\right) dy - C_{Y0} \cdot C_{Z0}\right] \tag{8}$$

$$A_s/2 = \arctan\left(C_{Z0}/C_{Y0}\right) \cdot R_a - C_{Y0} \cdot C_{Z0} \tag{9}$$

Where R_a is the radius of the curvature of the incoming cross-section, D_Z is the distance along the Z-axis direction between the origin coordinate and the center of the arc R_1, R_1 is the radius of the oval groove , H_p is the maximum thickness of the roll groove area.

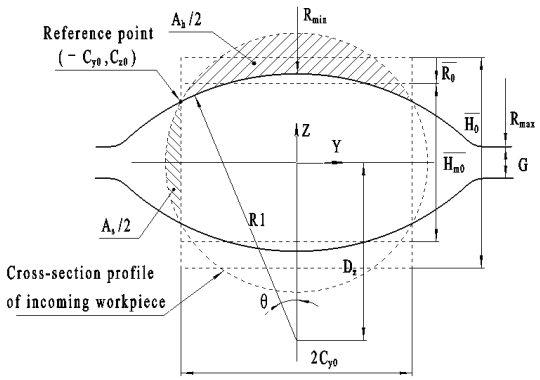


Figure 1. The parameter for the initial spread

2.2 The model of solving the critical point (C_Y, C_Z)

The critical point is defined as the point on the contact boundary at the exit section. The critical point (C_Y, C_Z) can be determined by solving two simultaneous circular equations.

According to the research of Y.Lee[9-12], R_s can be predicted as the linear interpolation of R_a and R_f (Fig.2) once maximum spread is known.

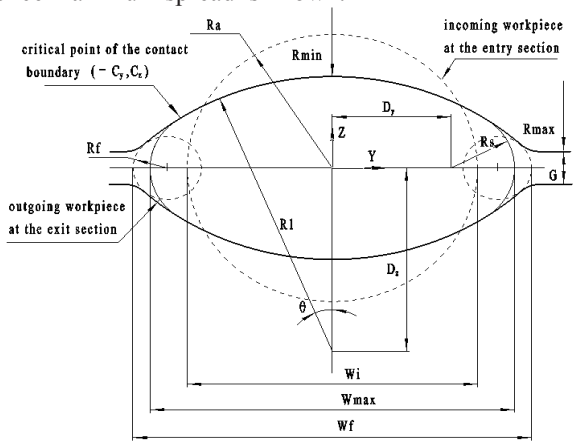


Figure 2. The parameter designation for the critical point

In round-oval pass rolling (Fig.2) , R_a and R_f may be given by

$$R_s = R_a \cdot w_t + R_f(1 - w_t) \tag{10}$$

$$R_f = \frac{R_1 \cdot H_p - \left(W_f^2 + H_p^2\right)/4}{2R_1 - W_f} \tag{11}$$

$$w_t = \frac{W_f - W_{\max}}{W_f - W_0} \tag{12}$$

$$W_{\max} = W_0 + \Delta b \tag{13}$$

where R_a is the radius of the curvature of the incoming cross-section, w_t is a weighting function, W_0 is the width of the incoming workpiece, W_f is the width of the roll groove area, Δb is the maximum spread of the outgoing workpiece.

The critical point (C_Y, C_Z) , at which the oval groove intersects the surface profile of outgoing workpiece, can be expressed explicitly in terms of a , b , c as

$$C_Z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{14}$$

where

$$a = 1 + \frac{D_Z^2}{D_Y^2} \tag{15}$$

$$b = 2D_Z - \frac{k_0 D_Z}{D_Y^2} \tag{16}$$

$$c = 2D_Z^2 - R_1^2 + \frac{k_0^2}{4D_Y^2} \tag{17}$$

$$k_0 = R_1^2 + D_Y^2 - R_s^2 - D_Z^2 \tag{18}$$

For the spread of the workpiece in rod rolling, the C_{Z0} is bigger than C_Z without the doubt. So, the criterion to determine the only effective value C_Z can be shown as

$$C_Z < C_{Z0} \tag{19}$$

$$C_{Z0} = \frac{R_1^2 - R_a^2 - D_Z^2}{2D_Z} \tag{20}$$

$$D_Y = \left(W_{\max} - 2R_s\right)/2 \tag{21}$$

$$D_Z = R_1 - H_p/2 \tag{22}$$

Once C_Z is determined, the C_Y can be obtained by

$$C_Y = \sqrt{R_1^2 - \left(C_Z + D_Z\right)^2} \tag{23}$$

2.3 The effective height and mean roll radius of outgoing workpiece

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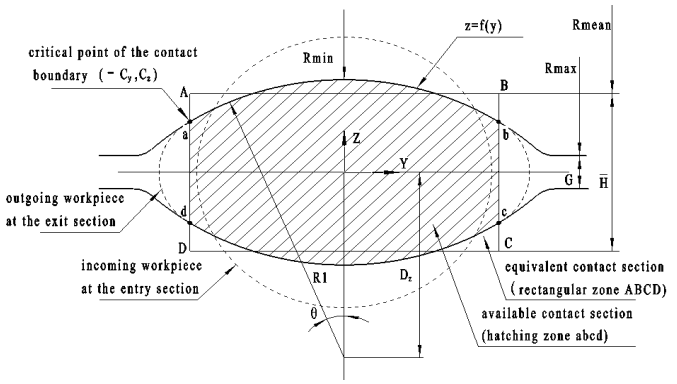


Figure 3. The parameter designation for the mean roll radius

In round-oval pass rolling, the curve equation of the oval groove is expressed as

$$z = f(y) = \sqrt{R_1^2 - y^2} - D_Z \tag{24}$$

Where R_1 is the radius of the oval groove, H_p is the thickness of the roll groove area (Fig.3).

The equivalent height of the workpiece in the oval pass is shown as

$$\overline{H_m} = \frac{2\int_{-C_Y}^{C_Y}\left(\sqrt{R_1^2 - y^2} - D_Z\right)dy}{2C_Y} = \frac{2\int_{0}^{C_Y}\left(\sqrt{R_1^2 - y^2} - D_Z\right)dy - 2C_Y \cdot D_Z}{C_Y} \tag{25}$$

Integrating the Eq. (25) yields

$$\overline{H_m}/2 = \frac{R_1^2(\sin 2\theta_0 + 2\theta_0)}{4C_Y} - D_Z \tag{26}$$

where θ_0 is shown as

$$\theta_0 = \arcsin(C_Y / R_1) \tag{27}$$

The mean roll radius of the oval pass and round pass may be given by

$$\overline{R} = R_{\max} + G/2 - \overline{H_m}/2 \tag{28}$$

2.4 Building the iterative model for spread prediction in Round-Oval-Round pass rolling

As can be seen in Eq.(1), the influence of cross-section profile of outgoing workpiece and the contact status was not been considered in this model, and only the influence of cross-section profile of incoming workpiece was considered. Consequently, it is not correct to substitute the effective height and mean roll radius of outgoing workpiece with $\overline{H_{m0}}$ and R_{m0} .The equivalent height and mean roll radius should be calculated by Eq.(26) and Eq.(28), respectively. So, the modified spread formula has been shown as follows

$$\Delta b = m \cdot \gamma \frac{\sqrt{\overline{R} \cdot \left(\overline{H_0} - \overline{H_m}\right)}}{W_0 + 0.5H_0} \frac{A_h}{A_0} \cdot W_0 \tag{29}$$

The iterative model for the spread prediction in alloy bar rolling can be shown as

$$\Delta b_i = m \cdot \gamma \frac{\sqrt{\overline{R_i} \cdot \left(\overline{H_0} - \overline{H_{mi}}\right)}}{W_0 + 0.5H_0} \frac{A_h}{A_0} \cdot W_0 \qquad (i \geq 0) \tag{30}$$

In round-oval pass rolling, the iterative formula for the equivalent height of outgoing workpiece can be shown as

$$\overline{H_{mi}} = \frac{R_1^2(\sin 2\theta_i + 2\theta_i)}{2C_{Yi}} - 2D_Z \qquad (i \geq 1) \tag{31}$$

$$\theta_i = \arcsin(C_{Yi} / R_1) \qquad (i \geq 1) \tag{32}$$

$$C_{Yi} = f_1(\Delta b_{i-1}) \qquad (i \geq 1) \tag{33}$$

The iterative model of mean roll radius can be shown as

$$\overline{R_i} = R_{\max} + G/2 - \overline{H_{mi}}/2 \qquad (i \geq 1) \tag{34}$$

Once the size of pass and cross-section profile of incoming workpiece is known, the initial spread can be calculated by Eq.(1).Then according to the prediction model of cross-section profile, the curvature radius of free surface R_s and the coordinates of critical point (C_Y, C_Z) can be obtained, moreover, the effective height of outgoing workpiece $\overline{H}$ and mean roll radius $\overline{R}$ can be obtained. So, according to the modified spread formula Eq.(29), a value of spread more correct can be calculated and another group of parameter about cross-section profile can be obtained. By many times iteration, an exact cross-section profile can be obtained.

3 The experimental results and discussion

The theoretical calculation and rolling experiments have been carried out according to the rolling schedule (Table.1) and pass schedule (Fig.4), which is always applied to Pomini Rolling Mills that belongs to BEIMAN Special Steel CO. LTD. The material of the workpiece is bearing steel(GCr15).The size of workpiece is 171mm diameter and 800mm length.

As shown in Table.2, the modified Shinokura and Takai's spread model was compared with other spread model, and it can be used to estimate the initial spread more precisely. As shown in Table.3, the predicting precision of this iterative model is satisfying enough. Therefore, it can be used to predict the maximum spread and design the profile of finishing groove in alloy bar rolling.

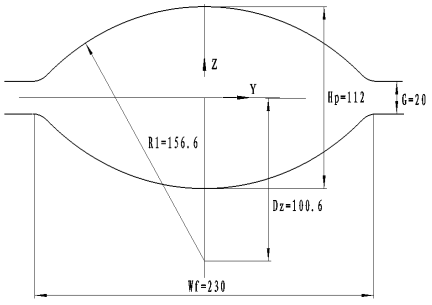


Figure 4. the size of the oval groove

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Table.1 Rolling schedule of the Pomini Rolling Mills

Parameter Pass	***Roll diameter (mm)***	***Roll speed (rpm)***	***Input speed (m/s)***	***Output speed (m/s)***	***Neutral diameter (mm)***	***Height of the Incoming workpiece (mm)***
Oval pass	730	8.1	0.22	0.28	663.6	171

Table.2 Initial spread by the different formula in alloy bar rolling

Spread model		***Smirnov***	***Wusatowski***	***Jiujin***	***Gubjin***	***Modified Shinokura***	***Experiments***
Value (mm)	Oval pass	18.5	14.6	15.1	25.3	30.20	34.8
	Round pass	14.2	11.5	11.6	18.7	20.15	24.0

Table.3 Comparison of iterative result and experimental results

Parameter Pass			R_s *(mm)*	(C_Y, C_Z) *(mm)*	$\overline{H}$ *(mm)*	$\overline{R}$ *(mm)*	$\overline{L}$ *(mm)*	Δb *(mm)*
Oval pass	Iterative result	Initial value	85.5	(77.95,35.25)	98.58	325.70	119.91	30.2
		1st iteration	48.65	(93.80,24.80)	92.09	328.92	129.07	32.51
		3rd iteration	45.91	(95.03,23.87)	91.52	329.24	129.83	32.72
		5th iteration	45.80	(95.12,23.73)	91.23	329.39	130.23	32.80
	Experimrntal data		43.50	(97.70, 21.80)	89.40	331.80	132.70	34.8
Round pass	Iterative result	Initial value	156.6	(48.60, 47.56)	123.66	309.17	130.08	20.15
		1st iteration	79.08	(58.46,34.73)	116.6	312.7	138.1	22.31
		3rd iteration	70.24	(59.63,32.84)	115.48	312.94	139.25	22.49
		5th iteration	68.37	(60.41,31.93)	114.86	313.17	140.23	22.65
	Experimental data		68.10	(56.40, 37.80)	113.20	317.90	141.80	24.80

4 Conclusions

1) Compared with other spread model, the modified Shinokura and Takai's spread model can be used to calculate the initial spread of iterative model in round-oval-round pass rolling. Moreover, an acceptable maximum spread can be obtained by this model if the required precision is not strict enough and the maximum spread is not big enough.

2) Based on the solution of critical point on the contact boundary, the equivalent height of outgoing workpiece and the mean roll radius can be calculated exactly, and the predicting precision of this iterative model can be improved greatly.

3) The influence of critical point and mean roll radius on the maximum spread has been considered in this model, so this iterative model has a high rate of convergence and its prediction precision is high enough.

4) Compared with the experimental results, the predicting precision of this iterative model is satisfying enough. Therefore, it can be used to predict the maximum spread and design the profile of finishing groove in alloy bar rolling.

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